

EE/CprE/SE 491 WEEKLY REPORT 1

09/08/2025 - 09/15/2025

Group number: 36

Project: Secure Wireless Aircraft Data Load

Client &/Advisor: Daniel Riechers & Mohammed Selim

Team Members/Role:

Anish Nag

Doyle Chism

Gurumanie Singh

Lincoln Hatlestad

Declan Creadon

Justin Rohde

○ Weekly Summary

Since it is the beginning of the semester, we discussed what roles and responsibilities we want to have throughout the semester. We chose responsibilities based on what area of the project each team member has knowledge in and what each team member desired. Our team also decided that we will break up into 2, 3-person pairs: One Cyber and One Non-Cyber. This allows for one team to focus on the cybersecurity aspect while the other focuses more on software. Lastly, we have met with our client advisor to discuss the project's purpose and get advice on direction. We have also decided when we will meet with our faculty advisor, so we have planned our meeting with him.

Past week accomplishments

Group: sdmay26-proj036

- Anish Nag: Worked on breaking down common industry standard PKI infrastructures and took notes on how we can possibly implement certain aspects of it into our project. This is important because in the case we have to build upon the standards and protocols already provided by Collins, will have some ideas to try and propose to the team.
- Gurumanie Singh: spent the week organizing and communicating with the team, Daniel Riechers (client advisor), and Mohammed Selim (faculty advisor) to find ideal meeting times. On the technical side, looked over one clear solution (a research paper) to our problem, summarized it and sent it to the group. Additionally, I also looked over more details about Raspberry Pi to see if it is a viable solution to our problem/project. Also looked into further details when it comes to speed of transfer, flexibility of Pi frameworks, and security needed when it comes to the data being transferred and also stored.
- Doyle Chism: Researched essays and RaspberryPi to find if there is a potential solution for wireless data transfer. Found a specific essay outlining a solution and we are asking Professor Selim(faculty advisor) and our client to see if this solution is viable for the RaspberryPi. We will then determine if we should proceed with it or make modifications based on Collins security clearances from open source projects.
- Justin Rohde: This week, read through the various security standards that our PoC sent us. Based on those standards, brainstormed solutions for our infrastructure that would tie into those standards to ensure that our solution is compliant.
- Lincoln Hatlestad: Researched ARINC 615A, a protocol our client advisor told us to learn about. Created some team organization documents including a task sheet, time sheets for each team member, meeting minute notebook, and project research notebook.
- Declan Creadon: Started research on the possibility of using a usrp based solution for this project. Read over some of the standards that our client advisor sent to us as well. Started to think about how we will encrypt the data and how we can securely transmit the data.

- **Pending issues** *(If applicable: Were there any unexpected complications? Please elaborate.)*
 - Gurumanie Singh: Read on some other solutions that may be out there that fit better with given situation, based on funding and other factors. There were some complications deciding between using Raspberry Pis and USRPs but we are working on it as a team by splitting the work. We have been productive when it comes to getting to the bottom of both solutions (Raspberry Pi vs USRPs) to see which works best for our project.
 - Doyle Chism: Some pending issues are that Collins has different security standards than our research paper has followed, so identifying changes to their solution is a complication we have to work through. Another issue is how we can configure a RaspberryPi in order to follow these security standards while still trying to properly complete a successful data transfer.
 - Anish Nag: Biggest issue is deciding between using microcontrollers or SDRs. They are both possible solutions so the aim is to find what works the most efficiently. We've decided to tackle this issue by breaking up into two groups: one group researches microcontrollers and the other group researches SDRs.
 - Justin Rohde: One pending issue is determining how the ARINC standards can be mapped onto Raspberry Pi or USRP implementations. Another pending issue is determining which standards are absolutely necessary to support the minimum viable prototype versus which can be deferred until later attempts.
- **Individual contributions** *(Creating this section is optional, but it is **Required to include the "Hours Worked for the Week" and their "Total Cumulative Hours" for the project for each member somewhere relevant in your report. Your individual weekly hours should be at a minimum of 6-8 hours for this course. So please manage your time well. Also, ensure that individual contributions support your claim to the weekly hours. Be honest with the reports.**)*

<u>Name</u>	<u>Individual Contributions</u> <i>(Quick list of contributions. This should be short.)</i>	<u>Hours this week</u>	<u>Hours cumulative</u>
Gurumanie Singh	● Researched Raspberry Pi and downsides (mainly) ● Contacting Prof Selim and Mr Riechers with regards to project specifications	7	20

Lincoln Hatlestad	● Researched ARINC 615A/665 ● Booked team meetings ● Meeting minutes	6	6
Anish Nag	● Researched raspberry pi microcontrollers versions ● Aggregated list of documentation ● Microsoft Learn: Certificates and PKI Infrastructure to get base understanding	6	6
Declan Creadon	● Started research on USRP based solutions ● Researched PKI and encryption ● Looked over the standards that were provided ● Started to think about the scope of cybersecurity in this project and what areas need to be covered.	6	6
Doyle Chism	● Discovered Research papers for potential solutions ● Researched RaspberryPi configuration ● Researched EAP-TLS and how it can be implemented within our RaspberryPi and environment ● Analyzed the ARINC 615A and how to implement the security into the RaspberryPi.	8	8
Justin Rohde	● Researched public key interface in wireless environments. ● Compared ARINC standards against potential Raspberry Pi and USRP implementations. ● Began outlining possible infrastructure designs that incorporate PKI.	5	5

○ **Comments and extended discussion** (Optional) (Feel free to discuss non-technical issues related

to your project).

Our group split into two sections, one group was tasked with researching RaspberryPi and one group was researching USRP's. Once we determine which hardware is best for our solution, then we will split up the tasks based on our experience to deliver the plan for the upcoming weeks.

- **Plans for the upcoming week** *(Please describe duties for the upcoming week for each member. What is(are) the task(s)?, Who will contribute to it? Be as concise as possible.)*

Our main task is researching Raspberry Pis and USRPs as options for our testing environment. We will split into two teams of three people to research either solution.

Team Raspberry Pis: Guru, Doyle, Lincoln

- Lincoln Hatlestad: Will research Raspberry Pis and other commercial off the shelf solution feasibility.
- Gurumanie Singh: Look into possible drawbacks and potential solutions when it comes to our project. Additionally, also look into add ons to the Raspberry Pi per our project requirements and see if it is still viable as a solution.
- Doyle Chism: Determine which process is the best option: RaspberryPi or USRP. Once we can figure out what is the best solution for our hardware then we can move forward on implementing the software solution. We can then deliver architectural design based on our solution moving forward. We would then need to acquire the hardware to begin the early stages of developing the software for the hardware.

Team USRPs: Anish, Declan, Justin

- Anish Nag: Research into the viability of using microcontrollers like a raspberry pi over USRPs like a SDR for creating a test environment. The first step will be to read documentation to understand the abilities and limitations of the microcontrollers. After that is done, will jot down ideas and architect possible solutions. These solutions will be diagramed and written down then proposed to the rest of the team so that we can come to a consensus on A)Does raspberry pi sound like the best option and B)which solution makes the most sense.
- Justin Rohde: Continue research on how ARINC standards (615A, 645, 811, DO-356a) and PKI requirements can be incorporated into a USRP-based approach. Compare feasibility of USRPs against Raspberry Pi for secure data load, focusing on compliance with aviation standards. Begin drafting diagrams of potential system architectures that integrate the standards into our prototype environment.
- Declan Creadon: Start narrowing down research on which encryption method will be the most efficient for our scenario. There could be one that is more secure but is too slow, need a faster one, etc. Additionally, will need to start looking into the speed and efficiency of USRPs and see if they are also fast enough.

- **Summary of weekly advisor meeting** *(If applicable/optional) (Provide a concise summary on the contents and progress made during the advisor meeting.)*

We discussed meeting and progress expectations with Professor Selim. Professor Selim prefers bi-weekly in-person meetings. We will have a report of our hours and progress for him at every meeting. We also discussed budget limitations. We then discussed possible hardware solutions for our project, and the feasibility of the research paper we planned to base our work on. Selim suggested researching USRPs(universal software radio peripheral) for our hardware solution. Selim also wanted us to vet the research paper some more to ensure reliability. Selim emphasized making a **doable** solution and then **upgrading** it if we can once we have a deliverable.